

BAB 13 AKSES GPS DAN OPENSTREETMAPS

A. Membuat Aplikasi Android

1. Buatlah *project* baru dengan detail sebagai berikut:

Application Name : Appx0c
Package Name : com.example.appx0c
Minimum SDK : API 22

2. Edit **AndroidManifest.xml**. Tambahkan kode berikut ini

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission
        android:name="android.permission.ACCESS_FINE_LOCATION"/>
    <uses-permission android:name="android.permission.INTERNET" />
    <uses-permission
        android:name="android.permission.ACCESS_NETWORK_STATE" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.App15"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category
                    android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

3. Pada **settings.gradle** tambahkan repositori **jitpack** seperti berikut ini

```
pluginManagement {
    repositories {
        google()
        mavenCentral()
        gradlePluginPortal()
    }
}
dependencyResolutionManagement {
    repositoriesMode.set(RepositoriesMode.FAIL_ON_PROJECT_REPOS)
    repositories {
        google()
        mavenCentral()
    }
}
```

```

        maven { url 'https://jitpack.io' }
    }
}
rootProject.name = "Appx0c"
include ':app'

```

4. Pada **build.gradle (project)** pastikan memiliki kode program seperti berikut ini

```

// Top-level build file where you can add configuration options common
to all sub-projects/modules.
plugins {
    id 'com.android.application' version '8.0.1' apply false
    id 'com.android.library' version '8.0.1' apply false
    id 'org.jetbrains.kotlin.android' version '1.8.20' apply false
}

```

5. Pada **build.gradle (module)** tambahkan dua buah pustaka, yaitu :

- **AirLocation** : digunakan untuk mempermudah mengakses fitur GPS di perangkat Android

```

plugins {
    alias(libs.plugins.android.application)
    alias(libs.plugins.kotlin.android)
}

android {
    namespace = "nugroho.benni.app15"
    compileSdk = 35

    defaultConfig {
        applicationId = "nugroho.benni.app15"
        minSdk = 25
        targetSdk = 35
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner =
"androidx.test.runner.AndroidJUnitRunner"
    }

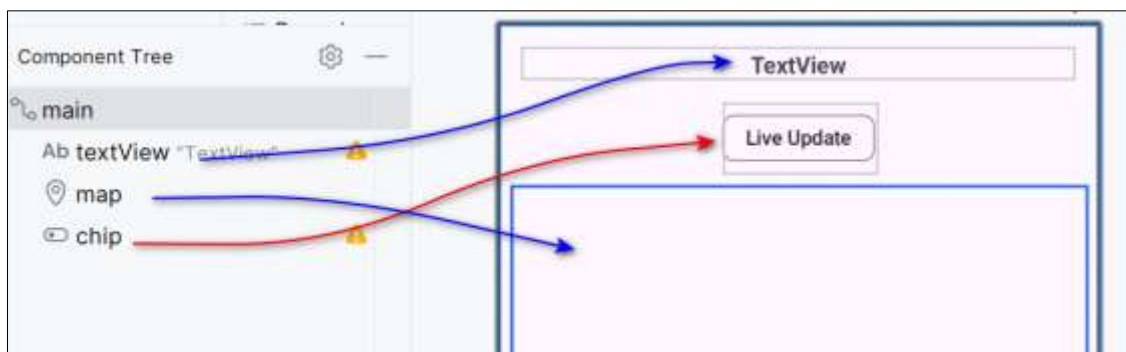
    buildFeatures{
        viewBinding = true
    }

    buildTypes {
        release {
            isMinifyEnabled = false
            proguardFiles(
                getDefaultProguardFile("proguard-android-optimize.txt"),
                "proguard-rules.pro"
            )
        }
    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
    kotlinOptions {
        jvmTarget = "11"
    }
}

```

```
dependencies {
    implementation("com.github.mumayank:AirLocation:2.5.2")
    implementation("org.osmdroid:osmdroid-android:6.1.10")
    implementation(libs.androidx.core.ktx)
    implementation(libs.androidx.appcompat)
    implementation(libs.material)
    implementation(libs.androidx.activity)
    implementation(libs.androidx.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.androidx.junit)
    androidTestImplementation(libs.androidx.espresso.core)
}
```

6. Buat layout antarmuka (UI) aplikasi dengan nama **activity_main.xml**. Untuk menampilkan maps di aplikasi digunakan view/widget **MapView**



Gambar : Layout antarmuka aplikasi pada activity_main.xml

Kode XML untuk antarmuka dalam **activity_main.xml** sebagai berikut

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:layout_marginTop="16dp"
        android:layout_marginEnd="16dp"
        android:text="TextView"
        android:textAlignment="center"
        android:textSize="16sp"
        android:textStyle="bold"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
```

```

<org.osmdroid.views.MapView
    android:id="@+id/map"
    android:layout_width="fill_parent"
    android:layout_height="0dp"
    android:layout_marginStart="8dp"
    android:layout_marginTop="8dp"
    android:layout_marginEnd="8dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/chip" />

<com.google.android.material.chip.Chip
    android:id="@+id/chip"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:checkable="true"
    android:text="Live Update"
    app:checkedIconVisible="true"
    app:chipIconVisible="true"
    app:layout_constraintEnd_toEndOf="@+id/textView"
    app:layout_constraintStart_toStartOf="@+id/textView"
    app:layout_constraintTop_toBottomOf="@+id/textView" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

7. Tuliskan kode program berikut pada *class file MainActivity.kt*

```

package nugroho.benni.app15

import android.content.Intent
import android.location.Location
import android.os.Bundle
import android.preference.PreferenceManager
import android.view.View
import android.widget.Toast
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
import mumayank.com.airlocationlibrary.AirLocation
import nugroho.benni.app15.databinding.ActivityMainBinding
import org.osmdroid.config.Configuration
import org.osmdroid.tileprovider.tilesource.TileSourceFactory
import org.osmdroid.util.GeoPoint
import org.osmdroid.views.MapView
import org.osmdroid.views.overlay.ItemizedIconOverlay
import org.osmdroid.views.overlay.ItemizedIconOverlay.OnItemGestureListener
import org.osmdroid.views.overlay.OverlayItem
import org.osmdroid.views.overlay.compass.CompassOverlay
import org.osmdroid.views.overlay.compass.InternalCompassOrientationProvider
import org.osmdroid.views.overlay.mylocation.GpsMyLocationProvider
import org.osmdroid.views.overlay.mylocation.MyLocationNewOverlay

class MainActivity : AppCompatActivity(), View.OnClickListener {

```

```
lateinit var b : ActivityMainBinding
lateinit var map : MapView
val arrayItemPos = ArrayList<OverlayItem>()
var updateLokasiSekaliSaja = false
var lat = 0.0;    var lng = 0.0;

fun drawMap() {
    //menambahkan kompas ke map
    val compass = CompassOverlay(applicationContext,
        InternalCompassOrientationProvider(applicationContext),
        map)
    compass.enableCompass()
    map.overlays.add(0, compass)

    //menghapus isi arrayItemPos bila sebelumnya sudah ada
    //biar markernya tidak redundant
    if(arrayItemPos.size > 0){
        for(i in 1 until map.overlays.size)
            map.overlays.removeAt(1)
        arrayItemPos.clear()
    }

    //menampilkan lokasi saya
    val gps = GpsMyLocationProvider(applicationContext)
    val me = MyLocationNewOverlay(gps, map)
    me.enableMyLocation()
    map.overlays.add(1, me)

    //zoom map sebesar 19
    val startPoint = GeoPoint(lat, lng)
    map.controller.setZoom(20.0)
    map.controller.animateTo(startPoint)

    //menambahkan titik lokasi tertentu yang akan ditandai
    arrayItemPos.add(OverlayItem(
        "Polinema Kampus Kediri 2", "Maskumambang",
        GeoPoint(-7.80203, 111.97982)))
    arrayItemPos.add(OverlayItem(
        "Polinema Kampus Kediri 1", "Semampir",
        GeoPoint(-7.801166, 112.008228)))

    //menambahkan marker ke map
    map.overlays.add(
        ItemizedIconOverlay<OverlayItem>(
            arrayItemPos,
            object : OnItemGestureListener<OverlayItem> {
                override fun onItemSingleTapUp(
                    index: Int, item: OverlayItem): Boolean {
                    Toast.makeText(applicationContext,
                        item.title, Toast.LENGTH_SHORT).show()
                    return true
                }

                override fun onItemLongPress(
                    index: Int, item: OverlayItem): Boolean {
                    Toast.makeText(applicationContext,
                        item.snippet, Toast.LENGTH_SHORT).show()
                    return true
                }
            })
    )
}
```

```
        }
    }, applicationContext
)
)
}

val airLocation = AirLocation(this, object: AirLocation.Callback {
    override fun onFailure(locationFailedEnum:
        AirLocation.LocationFailedEnum) {}

    override fun onSuccess(locations: ArrayList<Location>) {
        //menampilkan latitude longitude ke textview
        lat = locations[0].latitude
        lng = locations[0].longitude
        b.textView.text = "Latitude = "+lat.toString()+
            "\nLongitude = "+lng.toString()
    }
}, isLocationRequiredOnlyOneTime = updateLokasiSekaliSaja
)

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    enableEdgeToEdge()
    b = ActivityMainBinding.inflate(layoutInflater)
    setContentView(b.root)
    ViewCompat.setOnApplyWindowInsetsListener(
        findViewById(R.id.main)) { v, insets ->
        val systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars())
        v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom)
        insets
    }
    b.chip.setOnClickListener(this)

    //Penting! Diperlukan sharedPreferences default oleh osmdroid
    Configuration.getInstance().load(this,
        PreferenceManager.getDefaultSharedPreferences(this))

    //inisialisasi obyek map
    map = b.map
    map.setTileSource(TileSourceFactory.MAPNIK)
    map.setMultiTouchControls(true)
}

override fun onResume() {
    super.onResume()
    map.onResume()
}

override fun onPause() {
    super.onPause()
    map.onPause()
}

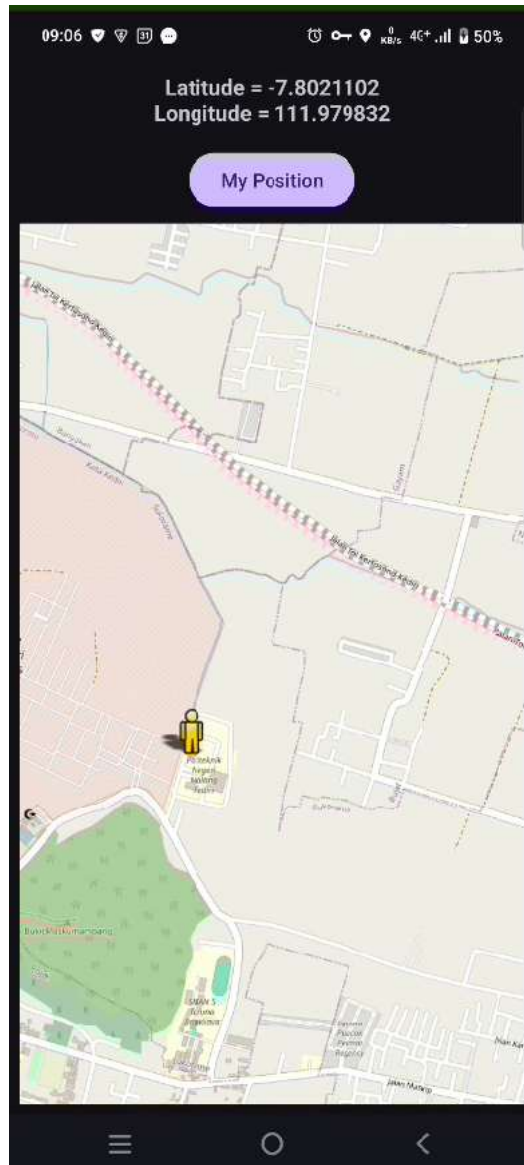
override fun onRequestPermissionsResult(
    requestCode: Int,
    permissions: Array<out String>,
```

```
grantResults: IntArray,  
deviceId: Int  
) {  
    super.onRequestPermissionsResult(requestCode,  
        permissions, grantResults, deviceId)  
    airLocation.onRequestPermissionsResult(requestCode,  
        permissions, grantResults)  
}  
  
override fun onActivityResult(  
    requestCode: Int,  
    resultCode: Int,  
    data: Intent?  
) {  
    super.onActivityResult(requestCode, resultCode, data)  
    airLocation.onActivityResult(requestCode, resultCode, data)  
}  
  
override fun onClick(p0: View?) {  
    when(p0?.id) {  
        R.id.chip -> {  
            b.chip.isChecked = updateLokasiSekaliSaja  
            updateLokasiSekaliSaja = updateLokasiSekaliSaja.not()  
            airLocation.start()  
            drawMap()  
        }  
    }  
}  
}
```

8.

B. Uji Coba Aplikasi

9. Jalankan aplikasi yang Anda buat untuk mengetahui lokasi Anda saat ini dan menunjukkan posisi latitude dan longitue anda seperti ditunjukkan oleh Gambar berikut



Gambar: Tampilan aplikasi setelah berjalan